AI Skincare Product Analyzer: PRJ1 Mid-Term Evaluation Report

Oct 22, 2026

This report presents the literature review, research gap, problem definition and proposed methodology for an AI Skincare Product Analyzer, covering all four mid-term components (25 marks in total).

Mid-term component	Marks	
Comprehensiveness of the Literature Review	10	Section 1
Research Gap	5	Section 2
Objective / Problem Definition	5	Section 3
Proposed Methodology (tools, techniques, methods, datasets)	5	Section 4

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1. Literature Review

Published work already shows that facial skin analysis, ingredient processing and product recommendation each work with AI, but few studies join all three and fewer still explain their recommendations. This section reviews ten studies, grouped by what they contribute to the proposed AI Skincare Product Analyzer, and then synthesises what they mean for the project.

All the content in this section is referred from the research papers listed in Section 5 (References).

1.1 Introduction

Choosing a skincare product is hard because skin type, visible concerns, sensitivities, allergies, preferences and budget differ from person to person. Ratings, advertising and popularity rankings ignore most of these factors. Three AI areas can help: computer vision for visible skin characteristics, natural language processing (NLP) for ingredient lists, and recommender systems for matching products to a user. This review treats the outputs as informational decision support. It does not treat image analysis as medical diagnosis, or ingredient associations as proof that a product works for an individual.

1.2 Scope and organisation

The reviewed studies fall into five groups:

- **A. Integrated image, ingredient and recommendation systems:** Lee et al. (2024).
- **B. Facial skin analysis:** Quattrini et al. (2022) and the smartphone acne detection and grading study (2022).
- **C. Knowledge-based and constraint-based recommendation:** Jannatin and Ratnasari (2025); Haloho and Liao (2026).
- **D. Content-based, clustering and review-based recommendation:** Wang et al. (2004); Putriany et al. (2019); Adebo (2020).
- **E. Surveys and comparative studies:** Vats et al. (2024); the IEEE review of cosmetic recommendation using deep learning.

For each study the entry gives the objective, method, key findings, limitations and relevance to this project. Where only an abstract was accessible, the entry says so.

1.3 Group A: Integrated systems

1.3.1 Lee et al. (2024): Deep learning-based skin care product recommendation

Objective: Build a personalised cosmetic recommender that uses both facial skin condition and ingredient information, instead of ratings or marketing text alone.

Method: A facial skin analyser uses deep learning to assess concerns such as pores, redness, acne, wrinkles, pigmentation and dark circles. A separate ingredient analyser treats an ingredient list as a sequence and applies a Transformer encoder with multilabel classification. A recommendation module combines both outputs. Product records were collected and refined from several sources.

Findings: The study shows that image-derived skin status and ingredient-based product information can be combined in one workflow, and that the approach gives reasonable recommendations for different skin problems.

Limitations: Performance depends on data coverage and quality, and on which skin conditions are represented. Predicted ingredient effects are estimates, not clinical evidence that a product will work for a given person.

Relevance: This is the closest study to the proposed project. It supports a modular design with separate image, ingredient and recommendation components. A student prototype can use simpler classifiers and transparent rule-based matching before attempting a Transformer.

1.4 Group B: Facial skin analysis

1.4.1 Quattrini et al. (2022): A deep learning-based facial acne classification system

Objective: Test whether deep learning can classify facial images as acne-affected or not.

Method: Facial images from the Flickr-Faces-HQ (FFHQ) dataset were annotated dermatologically. A semantic segmentation stage isolated relevant skin regions, and a deep classifier then separated acne from non-acne faces.

Findings: The final model reached an average F1 score of 60.84%. Automated acne classification is feasible, but the score shows that performance must be measured, not assumed.

Limitations: The study covers acne only and has no ingredient-aware recommendation step. The moderate F1 score leaves clear room for improvement.

Relevance: It shows the value of face-region segmentation before classification, and supports precision, recall, F1 and a confusion matrix as the evaluation set for the image module.

1.4.2 Automatic acne object detection and severity grading using smartphone images (2022)

Objective: Detect individual acne lesions in smartphone photographs and estimate severity, rather than labelling a whole face as acne or not.

Method: Faster R-CNN detects and classifies lesions, and LightGBM estimates severity from the lesion information. The dataset contains 1,572 labelled facial images captured with iOS and Android phones.

Findings: A lesion-level pipeline, where detected lesion types and counts feed a severity estimate, is more granular than whole-image classification.

Limitations: The system is specific to acne. It has no general skin profile, ingredient interpretation or recommendation step.

Relevance: It frames the choice between image classification and object detection. Classification is the realistic first step for this project; lesion detection is a possible extension if data and compute allow. It also shows that ordinary smartphone photos can be used as input.

1.5 Group C: Knowledge-based and constraint-based recommendation

1.5.1 Jannatin and Ratnasari (2025): Ontology and user preferences

Objective: Personalise skincare recommendations by representing the links between products, ingredients, skin types, concerns and user preferences in a structured knowledge base.

Method: The authors built an ontology with the Methontology framework and queried it with SPARQL on Apache Jena Fuseki. The knowledge base holds more than 3,800 products and 28,000 ingredient records. Ten respondents took part in a user-acceptance evaluation.

Findings: Explicit relationships between user needs, ingredients and products support structured and potentially explainable recommendations.

Limitations: Facial image analysis is not the main input, and ten respondents is too few to generalise from.

Relevance: It supports collecting skin type, concerns, sensitivity, allergies and budget through a questionnaire and matching them against structured product data. A relational database with clear rules is more feasible for a student prototype than a full ontology.

1.5.2 Haloho and Liao (2026): Three-stage hybrid recommendation for products and routines

Objective: Align user requirements, ingredient safety and product quality in one recommender that handles single products and multi-product routines. *Only the abstract was accessible; the chapter is behind a paywall.*

Method: Stage 1 applies constraint-based filtering. Stage 2 ranks the survivors by semantic similarity using a bi-encoder or a cross-encoder. Stage 3 applies a neural network that assesses product quality. For routines, an ingredient-compatibility component keeps product combinations safe while preserving diversity.

Findings: Single-product recommendation with the cross-encoder reached 84.68% requirement satisfaction with 15.8% catalogue coverage. Routine recommendation with the bi-encoder reached 84.22% satisfaction with 42.72% diversity. The authors conclude that constraint filtering provides the foundational accuracy, while the semantic and neural stages add diversity and quality.

Limitations: The abstract does not describe a facial image input. The evaluation measures requirement satisfaction rather than outcomes for real users, and the three-stage design is heavier to build and maintain than a rule-based one.

Relevance: It supports the project's constraints-first design, in which hard user constraints are applied before any ranking. Routine recommendation and ingredient-compatibility checks are sensible future work.

1.6 Group D: Content-based, clustering and review-based recommendation

1.6.1 Wang et al. (2004): A personalised recommender system for the cosmetic business

Objective: Explore personalisation and recommender methods in cosmetics retail.

Method: The paper is an early cosmetics-domain example of personalised recommendation. Content-based methods use product attributes and user preferences; collaborative methods use patterns in user and product interactions. *Check the full text before attributing a specific technique to the authors.*

Findings: It gives historical context and shows why recommendation is a separate technical component from image analysis.

Limitations: It predates deep-learning facial analysis. Classical methods also need product metadata and interaction data, which are sparse for new users and products.

Relevance: It gives a basis for explaining why the proposed system adds facial analysis and ingredient data to ordinary product matching.

1.6.2 Putriany et al. (2019): Item clustering with content-based filtering

Objective: Recommend skincare products that suit a user's skin profile by narrowing a large catalogue efficiently. *Only the abstract was readable.*

Method: Content-based filtering with K-means clustering groups similar products, and the clusters are used to calculate suggestions based on products the user already likes.

Findings: The system recommends products similar to ones the user likes, so users find products that may suit their skin.

Limitations: Recommendations start from products the user already likes, which limits new discovery. The abstract describes no image analysis, ingredient restrictions or budget handling.

Relevance: K-means clustering is a low-cost way to group products by category and ingredient profile, and to offer "similar products" as an optional extra.

1.6.3 Adebo (2020): An NLP approach to a skincare recommendation engine

Objective: Test how well text analysis of online reviews can drive skincare recommendations, to avoid dependence on explicit ratings.

Method: Over one million Reddit r/SkincareAddiction comments (18 months, 2018 to 2019) were matched against a 1,833-ingredient dictionary from Paula's Choice and 2,803 products scraped from Boots. VADER, extended with the SocialSent lexicon, converted sentiment into inferred ratings. The model is neural collaborative filtering (NCF) plus reviews: user and product embeddings with an LSTM (100 units) over review text, trained on an 80/20 split with mean squared error (MSE).

Findings: MSE was 0.30 for NCF plus reviews, against 0.32 for NCF, 0.39 for matrix factorisation and 0.66 for HFT. Most inferred sentiment was strongly positive.

Limitations: Ratings were inferred from unlabelled text by a lexicon, and the author notes that feature engineering limits generalisability. The model predicts ratings, not ingredient safety, and it has no image input or hard user constraints.

Relevance: It provides a worked method for building an ingredient dictionary, cleaning ingredient text and handling cold start. It also shows that review-based collaborative methods need large interaction data that a student project will not have.

1.7 Group E: Surveys and comparative studies

1.7.1 Vats, Hasija and Batra (2024): Comparative analysis of recommender approaches for skincare

Objective: Compare demographic, content-based, collaborative, hybrid and deep-learning recommenders for skincare. *Only the abstract of this four-page conference paper was accessible.*

Findings: It frames the choice of approach as a trade-off and shows that no single method fits every setting.

Limitations: It is short and comparative; it does not build or evaluate a new system.

Relevance: It supports choosing content-based filtering plus rules as the first implementation, since collaborative methods suffer from cold start when no user history exists.

1.7.2 A review on cosmetic product recommendation using deep learning (IEEE)

Objective: Survey recommenders that help users choose cosmetics suited to their skin. *Only the abstract was accessible; confirm the authors and year from the IEEE page.*

Scope: It covers face recognition, cosmetic product recognition, tag recommendation, CNN-based facial skin image classification and an examples-rules guided deep neural network. It analyses how user preferences, skin type, concerns and product attributes feed recommendation.

Relevance: It confirms that skin type, concerns and product attributes are the standard inputs, and that CNN-based facial skin classification is an established technique for the image module.

1.8 Comparative analysis

Study	Input	Main technique	Scale / reported result	Limitation for this project
Lee et al. (2024)	Face image, ingredient list	Deep skin analysis, Transformer ingredient model	Reliability reported per task	Data coverage; effects are estimates

Study	Input	Main technique	Scale / reported result	Limitation for this project
Quattrini et al. (2022)	Face image	Segmentation and deep classifier	F1 60.84% (acne)	Acne only; no recommendation
Acne detection (2022)	Smartphone photo	Faster R-CNN and LightGBM	1,572 labelled images	Acne only
Jannatin and Ratnasari (2025)	Questionnaire	Ontology, SPARQL	3,800+ products, 28,000+ ingredients; 10 respondents	No image input; small user test
Haloho and Liao (2026)	User requirements	Constraints, semantic ranking, neural quality score	84.68% satisfaction (single); 84.22% (routines)	No image input described; heavy to build
Wang et al. (2004)	Preferences, interactions	Content-based and collaborative	Not stated in this review	Predates deep learning
Putriany et al. (2019)	Liked products	Content-based filtering, K-means	Abstract only	Needs prior likes; no constraints
Adebo (2020)	Reddit reviews	NCF and LSTM, VADER	MSE 0.30 vs 0.39 (MF)	Rating prediction only; no image or safety
Vats et al. (2024)	None (comparison)	Review of five approach families	Abstract only	No new system
IEEE review	None (survey)	Survey of deep learning methods	Abstract only	No new system

1.9 Synthesis

Three research directions meet in the proposed system, and the reviewed work shows the state of each.

1. **Facial skin analysis works but is narrow.** Classification, segmentation and detection can estimate visible characteristics, yet the acne studies cover one condition and report moderate scores. Image quality, lighting and dataset diversity affect reliability.
2. **Ingredient information adds value but is uncertain.** Deep ingredient analysis and ontologies both link ingredients to concerns. Their outputs are associations, not guaranteed outcomes, and product data are often incomplete.
3. **Recommendation design decides whether results are usable.** Collaborative and review-based methods need interaction data. Constraint-first and content-based designs work when a new user supplies only a profile, which matches a student prototype.

One further study, outside the supplied list, is close to this project: Hansanie and Silva (2024) combine a CNN for acne severity with an ontology that uses skin type, concerns and allergy ingredients, and report 87.5% accuracy in a survey of 24 participants. It confirms that image, allergy and product data can be combined, and it relies on a small survey rather than a separate evaluation of each module.

These findings lead to four modules: image-quality checking and facial skin analysis; questionnaire-based user profiling; ingredient and product-data processing; and explainable, constraint-first recommendation. Published results will not transfer automatically, so the project must evaluate its own models and data.

1.10 Conclusion of the review

Facial skin analysis, ingredient processing and personalised recommendation have each been studied with machine learning and knowledge-based methods. Lee et al. is the closest integrated system, the acne studies supply image techniques, the ontology, constraint-based and hybrid studies supply knowledge-based and constraint-based matching, and the remaining studies supply the recommender background. The project should state uncertainty clearly and should not claim to diagnose conditions or guarantee product outcomes.

2. Research Gap

The literature does not leave integration unexplored: Lee et al. and Hansanie and Silva already join image analysis with product matching. The open problem is a reproducible, explainable system that treats user constraints as rules, reports the reliability of each module separately, and is evaluated end to end. The six gaps below are drawn from the limitations found in Section 1.

2.1 Gaps identified in the literature

#	Gap	Evidence in the reviewed work	Response in this project
G1	Model reliability varies by task, and ingredient effects are estimates	Ingredient-effect labels in Lee et al. were built largely from product marketing tags, with imbalanced classes and weaker test than validation scores (check the exact F1 values in the full text); acne classification reached F1 60.84% (Quattrini et al.)	Treat model outputs as uncertain estimates, report per-class metrics, show a confidence level, and avoid relying on predicted ingredient effects for safety decisions
G2	Image analysis covers few characteristics and depends on data quality	Acne dominates the image studies; Lee et al. list limited condition coverage; Quattrini et al. re-annotate FFHQ images	Support only characteristics with enough labelled data, audit class balance, and test results across image conditions such as lighting and skin tone
G3	Hard user constraints are rarely the first step of ranking	Adebo and Putriany et al. rank by similarity or predicted rating with no restriction filter; Jannatin and Ratnasari use preferences without images; Haloho and Liao use constraints but their abstract describes no image input	Apply budget, category and restricted-ingredient rules as eligibility filters before scoring, and measure the constraint violation rate

#	Gap	Evidence in the reviewed work	Response in this project
G4	Recommendations are seldom explained to ordinary users	Adebo relies on a neural rating model; Jannatin and Ratnasari give structure for explanation but show no user-facing reasons	Attach plain-language reasons to every recommendation, listing the matched concern, ingredient, category and budget
G5	Evaluation is inconsistent and often small	Metrics differ across studies (MSE, F1, requirement satisfaction); user tests used 10 and 24 people	Evaluate each module on its own metric, then the whole pipeline on predefined test personas, with a small usability study
G6	Complex designs are hard to reproduce, and claims can overreach	Transformer, neural-quality and ontology designs need large data or heavy tooling; ingredient presence does not prove suitability	Build a modular, rule-based baseline from open tools, and state plainly that output is informational and not medical advice

2.2 Consolidated research gap

The reviewed studies address parts of the problem. Lee et al. and Hansanie and Silva combine image analysis with product matching, Haloho and Liao show that constraint filtering gives strong accuracy, and Jannatin and Ratnasari show that structured knowledge supports personalisation. None of the reviewed studies reports one open, reproducible implementation that applies a user's allergy, budget and category constraints as hard rules, reports image-model reliability separately, explains each recommendation in plain language, and evaluates the whole pipeline against predefined test cases.

2.3 Research gap statement

The central gap is the practical implementation and systematic evaluation of a personalised, ingredient-aware and explainable skincare recommendation workflow. It combines estimates of visible facial skin characteristics with questionnaire-based user

requirements, while accounting for dataset limitations, model reliability and incomplete product information.

2.4 Research questions

1. **RQ1.** How reliably can a transfer-learning image classifier identify a small set of visible skin characteristics from facial images, and how does that reliability change with image conditions?
2. **RQ2.** How completely and consistently can public product and ingredient data be cleaned and standardised to support restricted-ingredient filtering?
3. **RQ3.** Does a constraints-first, weighted content-based recommender return relevant products while never violating explicit user constraints?
4. **RQ4.** Do plain-language explanations make the recommendations understandable to users?

2.5 Expected contribution

The contribution is a working, modular prototype with a documented evaluation, not a new learning algorithm. It shows which parts of the pipeline can be built reliably with open data and student-scale resources, and where reliability falls short.

3. Problem Definition and Objectives

This project builds an AI-based decision-support tool that matches skincare products to a person's visible skin characteristics, ingredient restrictions, preferences and budget, and explains why each product was chosen.

3.1 Problem statement

People differ in skin type, visible concerns, ingredient allergies, sensitivities, preferences and budget. Most product choices still rest on popularity, ratings, advertising or general descriptions, none of which account for an individual's requirements. Ingredient lists are long and technical, so comparing products on composition, price and intended use is slow and confusing.

The problem is therefore to develop an AI Skincare Product Analyzer that:

1. Analyses selected visible skin characteristics from a facial image;
2. Processes cosmetic ingredient lists and flags restricted ingredients;

3. Recommends and ranks products using the user's profile, ingredient restrictions, product category and budget; and
4. Explains each recommendation in plain language.

The system is an informational decision-support tool. It does not replace dermatological consultation and does not give medical diagnoses.

3.2 Aim

To design, implement and evaluate a modular prototype that combines computer vision, NLP and content-based recommendation to give personalised, explainable skincare product suggestions.

3.3 Objectives

The targets in the last column are proposed and will be confirmed after the dataset audit in Phase 2 of the plan.

ID	Objective	Measure of success (proposed)
O1	Build a facial image analysis module: preprocess images (resize, normalise, quality check) and classify a small set of visible characteristics, such as acne-like spots and redness, with a transfer-learning CNN	Macro-averaged F1 reported on a held-out test set, with precision, recall and a confusion matrix, beating a simple baseline; unusable images rejected before analysis
O2	Build a clean, structured product database from public and permitted sources (name, brand, category, price, ingredient list, intended use)	Duplicates removed; at least 1,000 products retained; ingredient lists present for at least 95% of retained records
O3	Build an ingredient analysis module that cleans, normalises and standardises ingredient names and matches them to a reference list	At least 95% of ingredient tokens mapped to a reference entry or flagged as unmatched, checked on a manually reviewed sample of 50 products
O4	Apply user constraints as rule-based filters before ranking: restricted ingredients, budget range and product category	No restricted ingredient in any recommendation where ingredient data are complete; violation rate reported

ID	Objective	Measure of success (proposed)
O5	Build a content-based recommender that scores eligible products with a weighted compatibility score and ranks them	Precision@5 and Recall@5 reported on predefined test personas with manually labelled relevant products
O6	Explain each recommendation in plain language	Every recommended product shows at least three reasons (for example concern match, preferred ingredient, category, price fit)
O7	Integrate all modules into one interactive application for input, upload, preferences and results	All workflow steps complete end to end in the prototype; input validation covers missing and invalid entries
O8	Evaluate the system and document limits and responsible use	Module-level and end-to-end results reported; usability score from at least 10 volunteers; images processed without permanent storage; limitations and future scope documented

These eight objectives consolidate the 18 objectives of the earlier draft: O1 covers 1 and 3; O2 covers 2 and 12; O3 covers 4; O4 covers 5, 7 and 8; O5 covers 6 and 9; O6 covers 10; O7 covers 11, 15 and 16; O8 covers 13, 14, 17 and 18.

3.4 Scope

In scope

- Estimating a few visible skin characteristics for which enough labelled data exist.
- A questionnaire for skin type, concerns, sensitivities, allergies, preferred category and budget.
- Ingredient parsing, matching and restriction flagging.
- Ranking, explanation and evaluation on test personas and a small user study.

Out of scope

- Medical diagnosis, treatment advice or claims of clinical effectiveness.
- Predicting skin type from images where no suitable labelled dataset exists; skin type is taken from the questionnaire.
- Live price tracking, purchasing and collaborative filtering that needs user history.

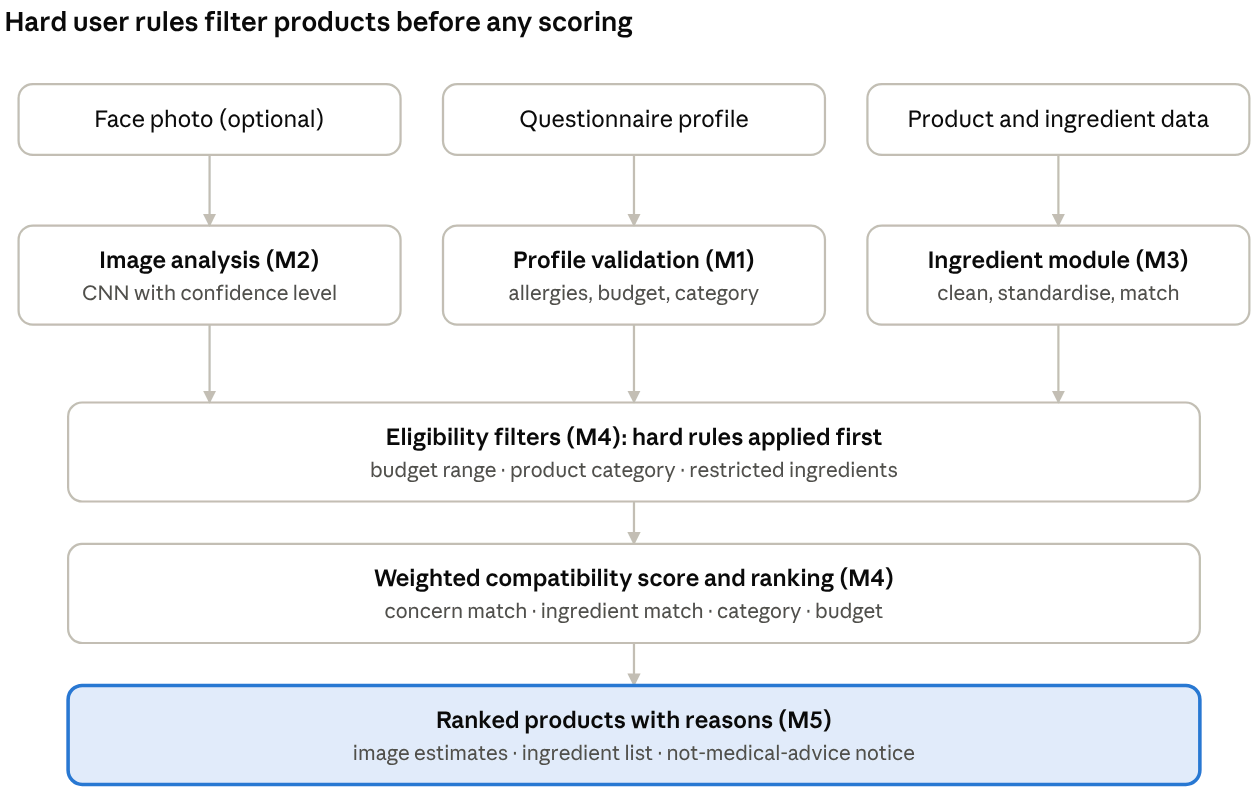
4. Proposed Methodology

The system is built as five connected modules. User constraints are applied as hard rules before any scoring, and every module is tested on its own before the whole pipeline is evaluated.

4.1 System overview

A user enters a profile and, optionally, a facial image. The image module estimates a few visible characteristics, the ingredient module standardises product ingredient lists, the filter removes products that break the user's rules, and the scorer ranks what is left. Each result is shown with its reasons.

The diagram below shows the flow of data from input to output.



system flow · 5 modules, hard filters before scoring

The image, profile and ingredient modules feed the filters, so a restricted ingredient or an out-of-budget price removes a product before it can be scored.

4.2 Datasets

No dataset is assumed to be downloaded or integrated yet. Each candidate is checked for label relevance, image quality, class balance, licence and permitted use before it is adopted.

Purpose	Candidate sources	Notes
Facial or skin images	Fitzpatrick17k; SCIN (Skin Condition Image Network); ACNE04 (acne grading and counting, Wu et al., ICCV 2019)	Confirm access terms and which labels exist. Fitzpatrick17k is a dermatology dataset with skin-tone annotation, not a skin-type dataset, so skin type comes from the questionnaire
Skincare products	Public skincare product datasets with ingredient lists (for example Kaggle skincare datasets); Open Beauty Facts; manufacturer or authorised retailer pages	Use only sources whose terms permit it. Fields: ID, name, brand, category, ingredient list, price, intended use, source URL
Ingredient reference	EU CosIng cosmetic ingredient database (INCI names and functions); a small hand-built synonym list	Functions are used for matching and explanation, not as proof of effect. Adebo used a scraped ingredient dictionary, so check licensing before reusing one
Testing data	About 20 predefined user personas written by the team; voluntary feedback from at least 10 volunteers	Consent and institutional rules apply when people are involved

4.3 Module design

4.3.1 Module 1: user input and validation

The form collects an optional face photo, skin type (oily, dry, combination, normal or unsure), visible concerns, allergies and ingredients to avoid, preferred category (cleanser, moisturiser, sunscreen and so on), a budget range and optional preferences such as fragrance-free. Entries are validated for missing and impossible values, such as a minimum budget above the maximum.

4.3.2 Module 2: facial image analysis

1. **Detect and crop:** Locate the face with OpenCV or MediaPipe and crop it. Reject images with no face or several faces.
2. **Quality check:** Reject images that are too blurred (low Laplacian variance), too dark or bright, or too small.
3. **Preprocess.** Resize to 224 by 224 pixels and normalise. Apply augmentation (flip, small rotation, brightness change) to training data only.
4. **Baseline:** Train a classical model (HOG and colour-histogram features with an SVM or logistic regression) so the deep model has something to beat.
5. **Main model:** Fine-tune a pretrained CNN (MobileNetV2, EfficientNet-B0 or ResNet50) by transfer learning: train the new head first, then unfreeze the upper layers. Use class-weighted loss for imbalanced labels.
6. **Split:** Stratified train, validation and test sets of 70, 15 and 15 percent, keeping images of the same person in one split where identifiers exist.
7. **Output:** Return each characteristic with a confidence value. Below a confidence threshold the result is shown as uncertain, and the profile falls back on the questionnaire.

The module will predict only characteristics for which a suitable labelled dataset is found. If no face dataset has usable labels, the project uses acne-focused data and says so.

4.3.3 Module 3: ingredient and product data processing

1. **Clean:** Lowercase, trim, remove stray symbols and percentages, and split the list on commas without breaking text inside brackets.
2. **Standardise:** Map synonyms to one reference name, for example aqua to water and perfum to fragrance, using the synonym list.
3. **Match:** Use exact matching first, then fuzzy matching (for example RapidFuzz with a high similarity threshold) for typing variants. Log everything left unmatched.
4. **Tag:** Attach documented function and category from the reference to each matched ingredient, and group ingredients into user-selectable families such as fragrance, essential oils, drying alcohols and sulphates.
5. **Flag completeness:** Mark each product with its ingredient-list completeness. When a user has restrictions, products with incomplete lists are excluded or clearly flagged, because a restriction cannot be checked on missing data.
6. **Clean the table:** Remove duplicates, fill or drop missing values, and standardise categories and prices.

The prototype uses dictionary and rule-based matching first. More complex NLP is considered only if matching accuracy is too low.

4.3.4 Module 4: constraint filtering and scoring

Step 1: eligibility filters. A product is removed if it is outside the budget range, is in the wrong category, or contains a restricted ingredient. These rules are never overridden by a high score.

Step 2: compatibility score. Each eligible product receives a weighted score:

```
S(p,u) = w_c\,C(p,u) + w_i\,I(p,u) + w_k\,K(p,u) + w_b\,B(p,u), \qquad w_c + w_i + w_k + w_b = 1
```

- **C** is the share of the user's concerns that the product's documented ingredient functions address.
- **I** is the overlap between the product's ingredients and the user's preferred ingredients.
- **K** is category relevance and intended use.
- **B** is budget suitability, higher when the price sits comfortably inside the range.

Starting weights of 0.35, 0.25, 0.20 and 0.20 are proposed. They will be tuned by grid search on the test personas, and a sensitivity check will show how much the ranking changes.

Step 3: optional similar products. TF-IDF cosine similarity on ingredient text, or K-means clustering as in Putriany et al., can suggest alternatives to a chosen product.

4.3.5 Module 5: explanation and output

Each recommended product lists the reasons behind its score, for example "fits your category and budget", "contains an ingredient you prefer" and "free of the ingredients you asked to avoid". The screen also shows the image-based estimates with their confidence, the ingredient list, any data-completeness warning, and a fixed notice that results are informational and not medical advice.

4.4 Tools and technologies

Tool	Purpose
Python	Main language for data, models and the app
Google Colab, Jupyter Notebook, VS Code	Experiments, training and evaluation

Tool	Purpose
OpenCV, MediaPipe	Face detection, cropping and quality checks
TensorFlow with Keras, or PyTorch	CNN transfer learning
scikit-learn	Baseline models, TF-IDF, K-means and metrics
Pandas, NumPy	Data cleaning and numerical work
NLTK or spaCy, RapidFuzz	Text cleaning and fuzzy ingredient matching
SQLite or CSV	Product, ingredient and test-persona storage
Streamlit	Interactive prototype
Matplotlib, Seaborn	Charts of data and results

The final choice of libraries depends on dataset compatibility and available compute.

4.5 Evaluation plan

Component	Method	Metrics
Image module	Held-out test set, compared with the classical baseline; results split by image condition and, where labels exist, skin-tone group	Accuracy, precision, recall, macro F1, confusion matrix, ROC-AUC
Data quality	Audit of the cleaned product table	Completeness, duplicate rate, share of ingredient tokens matched
Ingredient matching	Manual check of 50 sampled products	Match rate and error count
Constraint filtering	Run all test personas	Constraint violation rate (target zero where data are complete)
Recommender	Predefined personas with manually labelled relevant products	Precision@K, Recall@K and NDCG@K for K = 5; ablation of rules only versus rules plus image-derived concerns

Component	Method	Metrics
Explanations and usability	Voluntary user study of at least 10 people	System Usability Scale score; 1 to 5 rating of explanation clarity
Whole system	Scripted end-to-end run	Steps completed, input errors handled, response time

Where no ground-truth relevance labels exist, quality is judged on defined test cases and manual relevance checks, and the report makes no unsupported accuracy claims.

4.6 Work plan

Phase	Work	Output
1	Literature review, research gap, problem definition, methodology	mid-term
2	Data collection and audit: images, products, ingredient reference, licences	Chosen datasets and an audit note; targets in Section 3.3 confirmed
3	Facial image module: preprocessing, baseline, transfer learning	Trained model and test metrics
4	Product database and ingredient module	Clean product table and matching report
5	Constraint filters, scoring and explanations	Working recommender on test personas
6	Streamlit application and integration	End-to-end prototype
7	Evaluation, usability study, final report	Results, limitations and future scope

4.7 Ethics, privacy and limitations

- **Privacy:** Face images and preferences are used only for the analysis, processed in memory without permanent storage, and never shared. Participants give informed consent.
- **No medical claims:** Outputs are estimates and informational guidance. The interface carries a visible notice, and recommendations are never described as a diagnosis or a guarantee of safety or effectiveness.

- **Bias and data limits:** Lighting, camera quality and skin tone can change image results, so performance is reported by condition. Product ingredient lists may be incomplete or out of date, and ingredient presence alone does not show that a product is effective or safe for one person.
- **Licensing;** Dataset licences and website terms are checked before use, and nothing is scraped where terms forbid it.
- **Expected outcome:** A working prototype with documented strengths and limits, which can later be extended with more characteristics, wider product coverage and routine recommendation.

5. References

The content of this report is referred from the following reference papers.

1. J. Lee, H. Yoon, S. Kim, C. Lee, J. Lee, and S. Yoo, "Deep learning-based skin care product recommendation: A focus on cosmetic ingredient analysis and facial skin conditions," *J. Cosmet. Dermatol.*, vol. 23, no. 6, pp. 2066-2077, 2024, doi: 10.1111/jocd.16218.
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